① $$Z = \left(\frac{V-Nb}{N}\right)^N \left(\frac{m k_B T}{2\pi\hbar^2}\right)^{3N/2} e^{N^2a^2/Vk_BT}$$

a) $$F = -k_BT\ln Z = -k_BT\left[N\ln\left(\frac{V-Nb}{N}\right) + \frac{3N}{2}\ln\left(\frac{mk_BT}{2\pi\hbar^2}\right) + \frac{N^2a^2}{Vk_BT}\right]$$

$$E = \frac{\partial}{\partial\beta}(\beta F) = -\frac{\partial}{\partial\beta}\left(N\ln\left(\frac{V-Nb}{N}\right) + \frac{3N}{2}\ln\left(\frac{m}{2\pi\hbar^2\beta}\right) + \frac{\beta N^2a^2}{V}\right)$$

$$= \frac{-3N}{2}\,\frac{2\pi\hbar^2\beta}{m}\left(\frac{-m}{2\pi\hbar^2\beta^2}\right) - \frac{N^2a^2}{V}$$

$$E = \frac{3N}{2}k_BT - \frac{N^2a^2}{V}$$

$$P = -\left(\frac{\partial F}{\partial V}\right)_{T,N} = k_BTN\left[\frac{N}{V-Nb}\,\frac{1}{N} + \frac{(-1)Na^2}{V^2k_BT}\right]$$

$$P = \frac{Nk_BT}{V-Nb} - \frac{N^2a^2}{V^2}$$

b) b accounts for finite size of particles and Nb is the volume they consume

a accounts for interactions between molecules

② $\mathcal{H} = -BM \qquad M = \mu_0 \sum_i \sigma_i \qquad \sigma_i = \pm 1$

a) $Z_1 = \sum_{\sigma_i} e^{\beta B M}$, $Z_N = \prod_i \sum_{\sigma_i} e^{\beta B \mu_0 \sigma_i} = \left(2\cosh \frac{\mu_0 B}{k_B T}\right)^N$

$F = -k_B T \ln Z_N = -N k_B T \ln\left(2\cosh \frac{\mu_0 B}{k_B T}\right)$

b) $\langle M \rangle = N\mu_0 \langle \sigma_i \rangle$

$\langle M \rangle = N\mu_0 \dfrac{\sum_i \sigma_i e^{\beta B \mu_0 \sigma_i}}{Z} = \dfrac{N\mu_0 \, 2\sinh(\beta B \mu_0)}{2\cosh(\beta B \mu_0)}$

$= N\mu_0 \tanh(\beta B \mu_0)$

High-T $\frac{B\mu_0}{k_B T} << 1 \qquad x = \frac{B\mu_0}{k_B T} \qquad \tanh(x) \sim x$

so $\langle M \rangle = N\beta B \mu_0^2$

Low-T $x >> 1 \qquad \tanh(x) \sim 1 \qquad \langle M \rangle = N\mu_0$

c) $S = -\left(\frac{\partial F}{\partial T}\right)_B = N k_B \ln(2\cosh x) + N k_B T \frac{\sinh x}{\cosh x}\left(\frac{-\mu_0 B}{k_B T^2}\right)$

$S = N k_B \left[\ln(2\cosh x) - x \tanh x\right]$

if S is constant, then x is constant

so $x_1 = x_2 \qquad \frac{\mu_0 B_1}{k_B T_1} = \frac{\mu_0 B_2}{k_B T_2} \qquad T_2 = \frac{B_2}{B_1} T_1$

(3) a) $\epsilon(k) = \frac{\hbar^2 k^2}{2m}$

$$N = \frac{L}{2\pi}\int_{-k_F}^{k_F} dk = \frac{L k_F}{\pi} \qquad k_F = \frac{N\pi}{L}$$

$$\varepsilon_F = \frac{\hbar^2 k_F^2}{2m} = \frac{\hbar^2 N^2 \pi^2}{2mL^2}$$

b) $$\frac{\langle \varepsilon \rangle}{N} = \frac{L}{2\pi N}\int_{-k_F}^{k_F} dk \, \frac{\hbar^2 k^2}{2m} = \frac{L\hbar^2}{4\pi m N}\frac{2}{3}k_F^3 = \frac{L\hbar^2}{6m\pi N}k_F^3$$

$$= \frac{L}{3\pi N}\varepsilon_F k_F = \frac{N\pi}{NL}\frac{L}{3\pi}\varepsilon_F = \frac{\varepsilon_F}{3}$$

c) $\langle \varepsilon \rangle = \frac{\langle p \rangle^2}{2m} \qquad \langle p \rangle = \sqrt{2m\langle \varepsilon \rangle}$

$$\langle p \rangle = \sqrt{2m\frac{\varepsilon_F}{3}} = \sqrt{\frac{2m}{3}\frac{\hbar^2 k_F^2}{2m}} = \sqrt{\frac{p_F^2}{3}} = \frac{p_F}{\sqrt{3}}$$

(4) $\mathcal{H} = \frac{-\hbar^2}{2I}\frac{d^2}{d\theta^2}$ $\quad 0 \le \theta < 2\pi$

a) $\mathcal{H}\psi = E\psi$ $\quad$ use $\psi = e^{in\theta}$

$\frac{-\hbar^2}{2I}(-n^2)\psi = E\psi$ $\quad E = \frac{\hbar^2 n^2}{2I}$ $\quad n = 0, \pm 1, \pm 2, \ldots$

b) $Z = \sum_n e^{\frac{-\hbar^2 n^2}{2Ik_BT}}$ $\quad$ low-T: $\frac{\hbar^2}{2Ik_BT} >> 1$

$x = \frac{\hbar^2}{2Ik_BT}$ $\quad x >> 1$ $\quad Z = \underset{n=0}{1} + \underset{n=1}{e^{-x}} + \underset{n=-1}{e^{-x}}$ $\quad$ only keep lowest order terms since $x >> 1$

$\cancel{Z = 1 + e^{-2x}}$ $\quad Z = 1 + 2e^{-x}$ for low-T

high-T $\quad x << 1$ $\quad$ can't expand

$Z = \sum_n e^{-xn^2}$ $\quad$ let $y^2 = xn^2$

$Z = \sqrt{\frac{2Ik_BT}{\hbar^2}}\int_{-\infty}^{\infty} e^{-y^2}dy = \sqrt{\frac{2Ik_BT\pi}{\hbar^2}}$

c) low-T: $x >> 1$ $\quad F = -k_BT\ln(1 + 2e^{-x}) \underset{\substack{\uparrow \\ \text{expansion} \\ \text{of } \ln}}{\simeq} -2k_BTe^{-x}$

$E = \frac{\partial}{\partial\beta}(\beta F) = \frac{\partial}{\partial\beta}(-2k_BTe^{-x})$

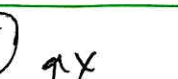
$E = \frac{\hbar^2}{I}e^{-\beta\hbar^2/2I}$ $\quad C_V = \frac{\partial E}{\partial T} = \frac{\hbar^4}{4I^2k_BT^2}e^{-\beta\hbar^2/2I}$

high-T: $x << 1$ $\quad F = -\frac{k_BT}{2}\ln\left(\frac{2I\pi}{\beta\hbar^2}\right)$

$E = \frac{\partial}{\partial\beta}(\beta F) = -\frac{1}{2}\left(\frac{\beta\hbar^2}{2I\pi}\right)\left(\frac{-2I\pi}{\hbar^2\beta^2}\right) = \frac{1}{2\beta} = \frac{1}{2}k_BT$

$C_V = \frac{\partial E}{\partial T} = \frac{k_B}{2}$

(5)



$L = (N_+ + N_-)$ total length

$x = (N_+ - N_-)a$ height above 0

$N = N_+ + N_- \rightarrow N_+ = N - N_-$

so $x = (N - 2N_-)a$ now call $N_- = n$

$x = (N - 2n)a$

a) $U = mgx$

$$Z = \sum_{n=0}^{N} \underbrace{\frac{N!}{N_+!N_-!}}_{\binom{N}{n}} e^{-\beta U} = \sum_{n=0}^{N} \binom{N}{n} e^{-\beta mgNa + 2\beta mgna}$$

$\binom{N}{n}$ accounts for all combinations

$$Z = e^{-\beta mgNa} \sum_{n=0}^{N} \binom{N}{n} q^N = e^{-\beta mgNa}(1+q)^N \quad \text{where } q = e^{2mga\beta}$$

binomial expansion from hint

$$Z = e^{-\beta mgNa}(1+e^{2\beta mgNa}) = \left(e^{-\beta mga}\right)^N (1+e^{2\beta mga})^N$$

$$= \left(e^{-\beta mga} + e^{\beta mga}\right)^N = \left(2\cosh(\beta mga)\right)^N$$

b) $\langle x \rangle = (N - 2\langle n \rangle)a$

$$\langle n \rangle = \sum_{n=0}^{N} \binom{N}{n} n e^{-\beta U}/Z \quad \text{let } \underline{y = mga\beta}$$

$$\langle n \rangle = \frac{1}{Z} e^{-Ny} \sum_{n=0}^{N} \binom{N}{n} n e^{2ny}$$

Now use the fact that $ne^{2yn} = \frac{1}{2}\frac{\partial}{\partial y} e^{2yn}$

then $$\langle n \rangle = \frac{e^{-yN}}{Z} \frac{1}{2}\frac{\partial}{\partial y} \sum_{n=0}^{N} \binom{N}{n} e^{2yn} =$$

$$= \frac{e^{-yN}}{Z} \frac{1}{2}\frac{\partial}{\partial y}(1+e^{2y})^N$$

(5) cont...

$$\langle n\rangle = \frac{e^{-yN}}{[2\cosh y]^N}\,\frac{1}{2}N\left(1+e^{2y}\right)^{N-1}2e^{2y} =$$

$$= \frac{N e^{-y} e^{-y(N-1)} e^{2y}\left(1+e^{2y}\right)^{N-1}}{[2\cosh(y)]^N}$$

$$= \frac{Ne^{y}\left(e^{-y}+e^{y}\right)^{N-1}}{(2\cosh(y))^N} = \frac{N e^{y}}{2\cosh y} = \frac{N e^{y}}{e^{y}+e^{-y}}$$

$$\langle x\rangle = (N-2\langle n\rangle)a = \left(N - \frac{Ne^{y}}{\cosh y}\right)a$$

c) $y \ll 1$ $\quad \langle x\rangle = Na\left(1-\frac{e^{y}}{\cosh y}\right)$ $\quad e^{y}\sim 1+y$

$\cosh y \sim 1+\frac{y^2}{2}$

$$\langle x\rangle = Na\left(1-\frac{1}{1+\frac{y^2}{2}}-\frac{y}{1+\frac{y^2}{2}}\right) = Na\left(\frac{2+y^2-2-2y}{2+y^2}\right)$$

$$= -Nay\left(\frac{2-y}{2+y^2}\right) \simeq -Nay$$ since y is small

so $-Na\frac{mga}{k_BT} = \langle x\rangle$ or $mg = -\frac{k_BT}{Na^2}\langle x\rangle$

which is the form of Hooke's Law with

$$k = \frac{k_BT}{Na^2}$$